SENTIMENT CLASSIFIER BIAS AUDIT

Evaluating Performance Across Demographic Subgroups

Text Analytics Term Project Report

Prepared as a reproducible educational prototype

Project scope: A baseline sentiment classifier is trained and evaluated, then its error patterns are compared across illustrative demographic and writing-style subgroups.

Data note: The included dataset is synthetic. Metrics demonstrate the tool workflow only and must not be interpreted as real-world demographic findings.

1. Abstract

This project presents a practical bias-auditing workflow for a binary sentiment classifier. Text is transformed into TF–IDF features and classified with Logistic Regression. Performance is summarized globally and separately for each subgroup using accuracy, positive-class precision, recall and F1, false-positive rate (FPR), false-negative rate (FNR), and positive prediction rate. A configurable disparity summary highlights metrics that exceed screening thresholds. The workflow also exports tables and plots to support review.

2. Problem Statement

Sentiment classifiers can perform differently across subgroups because of differences in language patterns, class balance, data quality, or model behavior. Reporting only a single aggregate score can conceal such differences. The project therefore builds a lightweight audit tool that compares performance across subgroup labels and makes possible disparities visible for further investigation.

3. Objectives

• Prepare a labeled text dataset for sentiment classification.

• Train a reproducible TF–IDF + Logistic Regression baseline.

• Compute overall and subgroup-level classification metrics.

• Compare error rates and flag gaps using configurable thresholds.

• Export tables, plots, and examples of misclassified text for inspection.

4. Dataset

The notebook creates 239 synthetic text records with a binary sentiment label (negative/positive), an illustrative gender-group field, and an illustrative writing-style field. Text examples are generated from positive, negative, and ambiguous neutral templates. A stratified 75/25 train/test split is used. The test set contains 60 records. Synthetic categories are used only to demonstrate the method and are not claims about real demographic language.

5. Methodology

• **Preprocessing:** Text is lowercased and tokenized by scikit-learn's TF–IDF vectorizer; unigrams and bigrams are used.

• **Model:** Logistic Regression with a maximum of 1,000 iterations.

• **Evaluation:** Overall metrics and subgroup metrics are computed on the held-out test split.

• **FPR:** FP / (FP + TN), the share of truly negative examples predicted positive.

• **FNR:** FN / (FN + TP), the share of truly positive examples predicted negative.

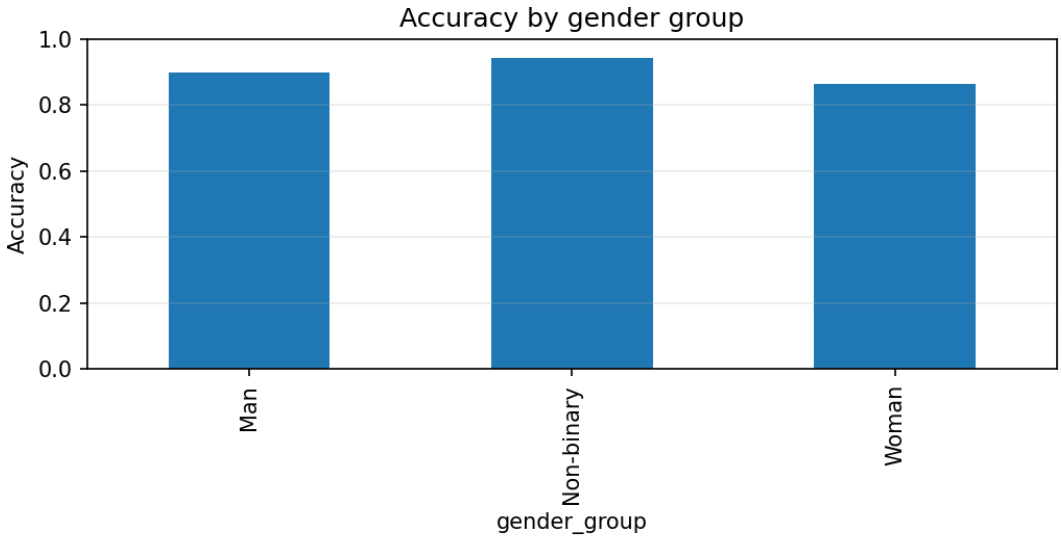
• **Disparity screening:** Absolute max-minus-min gaps are compared with configurable 0.10 thresholds. A flag means review is recommended, not that unfairness has been proven.

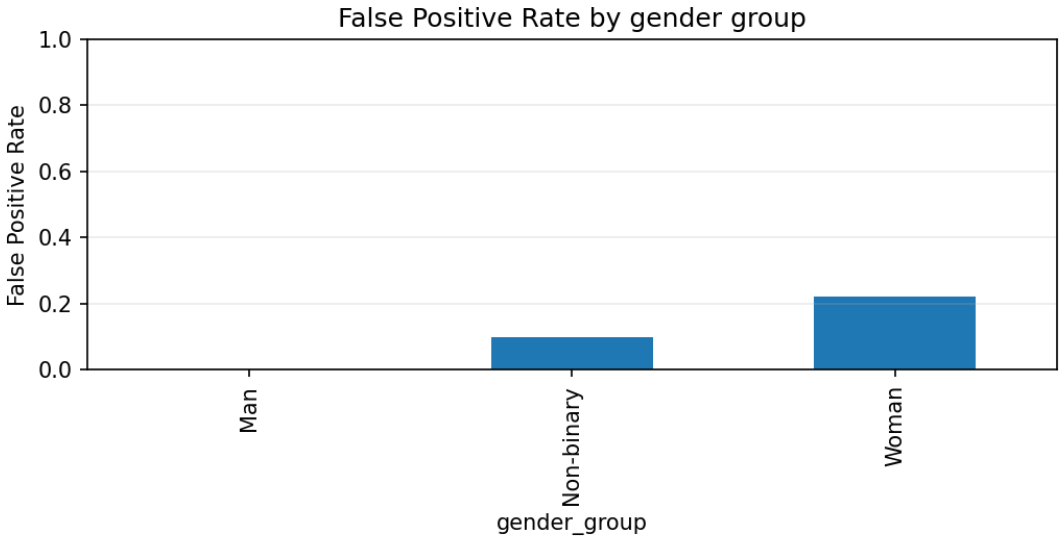
6. Results

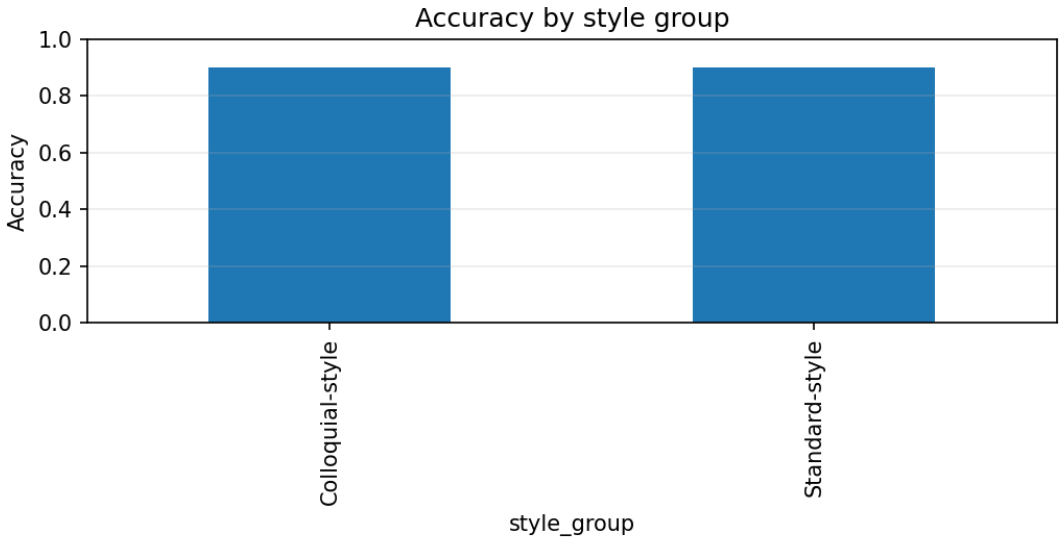
Metric	Observed value
Test rows	60
Accuracy	0.900

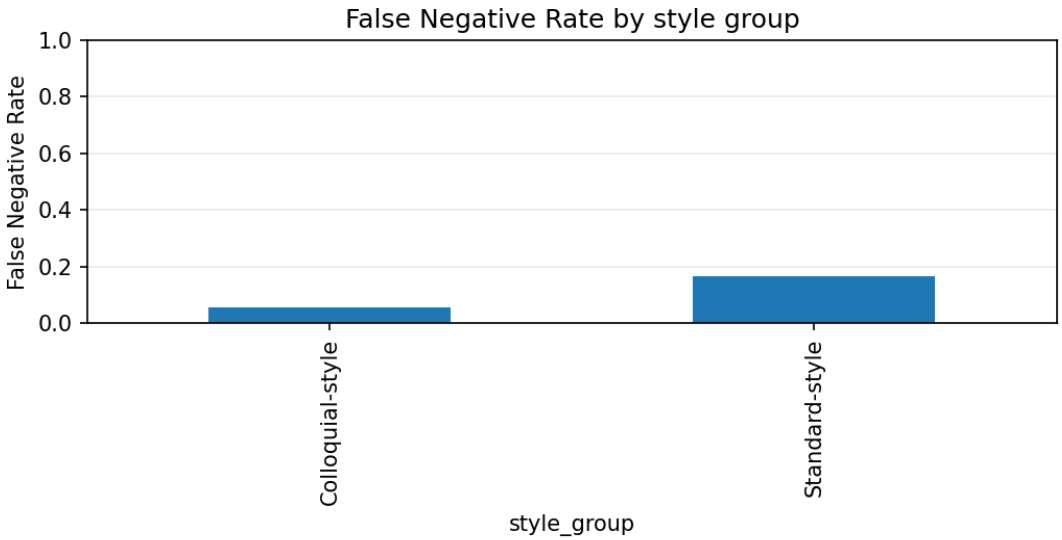
Positive precision	0.900
Positive recall	0.900
Positive F1	0.900
False-positive rate	0.100
False-negative rate	0.100

The notebook produces subgroup tables for gender_group and style_group. These tables include subgroup sample counts, error rates, and prediction rates. Small groups should be interpreted cautiously because estimates can be unstable. See the accompanying Output_Screenshots.pdf for metric charts and example output snapshots.









7. Discussion

The audit is designed to encourage a move beyond aggregate accuracy. A difference in FPR may be important when false positive sentiment labels have a meaningful downstream impact; a difference in FNR may matter when positive sentiment is being missed. The appropriate metric depends on the intended use. A disparity flag is a prompt for analysis, not a standalone fairness verdict. Sampling variation, label noise, domain shift, and subgroup size can all affect the observed results.

8. Limitations and Ethical Considerations

• The dataset is synthetic, template-based, and small compared with a production corpus.

• The subgroup attributes are illustrative; demographic attributes should not be inferred from text or names.

• Binary sentiment labels do not represent the full complexity of human sentiment or sarcasm.

• A 0.10 gap threshold is a configurable screening heuristic, not a universal legal or ethical standard.

• Before deployment, use representative and appropriately licensed data, review consent and privacy, report uncertainty, and involve domain experts and affected stakeholders.

9. Future Enhancements

Recommended improvements include bootstrap confidence intervals, subgroup-aware sample-size warnings, calibration analysis, threshold sweeps, multilingual testing, more robust datasets, human review of errors, comparison across model families, and a dashboard for interactive exploration.

10. Conclusion

The project provides a reproducible prototype for auditing a sentiment classifier across subgroups. It trains a baseline model, computes overall and subgroup metrics, visualizes differences, and exports results. Its main contribution is a transparent workflow for discovering questions that deserve deeper investigation. Any real-world fairness claim requires appropriate data, uncertainty analysis, and contextual review.

Appendix: Files

sentiment_bias_audit.ipynb — complete notebook; bias_audit.py — script alternative; outputs/ — generated CSV tables and PNG plots; Output_Screenshots.pdf — output snapshots; Demo_Video.mp4 — short walkthrough.